Compound epithelium is made of more than one layer (multi-layered) of cells and thus has a limited role in secretion and absorption(Figure 3.4). The compound epithelia may be stratified and transitional. Their main function is to provide protection against chemical and mechanical stresses. They cover the dry surface of the skin, the moist surface of buccal cavity, pharynx, inner lining of ducts of salivary glands and of pancreatic ducts. There are four types of compound epithelium namely, stratified squamous epithelium, cuboidal epithelium, columnar epithelium and transitional epithelium. **Stratified squamous epithelium** is of two types called **keratinized type** which forms the dry epidermis of the skin and the **non keratinized type** forms the moist lining of the oesophagus, mouth, conjunctiva of the eyes and vagina. **Stratified cuboidal epithelium** mostly found in the ducts of sweat glands and mammary glands. **Stratified columnar epithelium** has limited distribution in the body, found around the lumen of the pharynx, male urethra and lining of some glandular ducts. **Transitional Epithelium** is found lining the ureters, urinary bladder and part of the urethra. This epithelium allows stretching and is protective in function.

All cells of the epithelium are held together with little intercellular material. In most of the animal tissues, specialized junctions provide both structural and functional links between its individual cells. Three types of cell junctions are found in the epithelium and other tissues. These are called as tight, adhering and gap junctions. **Tight junctions** help to stop substances from leaking across a tissue. **Adhering junctions** perform cementing to keep neighbouring cells together. **Gap junctions** facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells, for rapid transfer of ions, small molecules and sometimes big molecules.

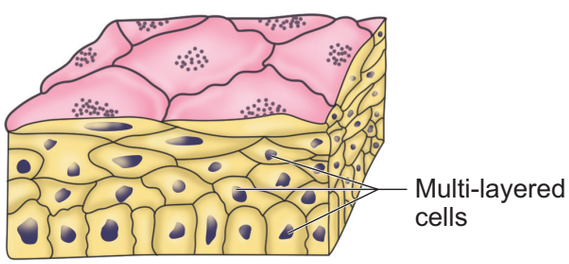


Figure 3.4 Compound Epithelium

Stratified epithelia are "built" for protection or to resist abrasion. What are the simple epithelia better at?

3.3 Connective Tissue

Connective tissue develops from the **mesoderm** and is widely distributed in the body. There are three main classes namely Loose connective tissue, Dense connective tissue and Specialized connective tissue. Major functions of connective tissues are **binding, support, protection, insulation** and **transportation**

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Connective Tissues

Loose Connective Tissues	Dense Connective Tissues	Specialised Connective Tissues
1. Areolar Tissue	1. Dense Regular	1. Cartilage
2. Adipose Tissue	2. Dense Irregular	2. Bone
3. Reticular Tissue	3. Elastic	3. Blood

1. What type of connective tissue is damaged when one get cut on his index finger accidently?
2. The stored lipids are in the form of adipose tissue. Are they coloured? why?

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